



M.Arch Javier Martínez Gaxiola / JAMAGAX

Senior Computational Design Director | CNC & Digital Fabrication Expert

Tulum, Quintana Roo — Mexico

PROFESSIONAL PROFILE

Director of Jamgax Studio and Founder of Dimensión N with over 20 years of experience bridging the gap between algorithmic design and industrial manufacturing. I specialize in translating complex, non-standard geometry into precise, buildable reality using **CNC, Advanced Parametric Logic, and Reality Capture pipelines.**

Former **Fab Lab Manager at Azulik** and Computational Lead for **Mayan Warrior**, I combine the precision of an Industrial Designer with the strategic vision of a Project Manager. Author of *La Geométrica Encyclopedia* and pioneer of Grasshopper education in Latin America.

PROFESSIONAL EXPERIENCE

Jamagax Studio — Tulum, MX

Founder & Lead Computational Designer | 2025 – Present

- Developing AI-assisted workflows using LLM architectures and Model Context Protocol (MCP) to automate project documentation and design pipelines.
 - Reality Capture & Optimization: Expert workflows for Drone Photogrammetry and 3D Scanning — specializing in repairing complex meshes from Point Clouds and Gaussian Splatting into clean, manufacturable geometry.
 - Designing web-based 3D digital twins (Three.js/GLSL) for interactive project visualization.
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Mayan Warrior — Burning Man | Remote & Black Rock City

Computational Design Lead & CNC Specialist | 2024 – 2025

- Led the computational strategy for the massive 4×4×4m aluminum head sculpture.
 - Managed the rationalization, nesting, and G-code logic for **1,500+ unique CNC-machined parts.**
 - Developed custom Python/Grasshopper scripts for automated labeling and assembly sequencing.
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Piegatto — Guatemala / Remote

Computational Specialist (Large-Scale 3D Printing) | 2025

- Developed parametric workflows for large-format 3D printing of furniture and lighting.
 - Optimized mesh topology for rapid additive manufacturing, reducing print time and material waste.
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Roth Architecture — Azulik | Tulum, MX

Fab Lab Manager & Production Director | 2023

- Orchestrated operations for the Digital Fabrication Department, coordinating workflows across KUKA Robotics, Industrial Waterjets, and CNC routers.
 - Strategic Planning: Served on the Board of Directors to align design vision with manufacturing capabilities.
 - EV Project Lead: Directed the manufacturing of large-scale MDF molds and high-density exotic wood components for the Azulik Electric Vehicle.
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FI Interiors — Orbus | Chicago, IL / Remote

Senior Computational Designer | 2021 – 2022

- Engineered parametric systems for complex trade show displays and retail interiors.
 - Automated the generation of shop drawings and cut-files, reducing manual drafting time by ~40%.
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Parametric Art & Fabrication | Mexico / Global

Lead Computational Specialist | 2017 – 2020

- **Xochipeople — Burning Man (2017):** Lead parametric design for a large-scale structural sculpture.
 - **El Excusado — Yoshua Okón & Santiago Sierra (2017):** Executed parametric reticulation and 1:1 scale 3D printing for international museum-grade art pieces.
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Bioclimatic Data Consulting | Mexico / Chile

Lead Environmental Specialist | 2014 – 2017

- Developed computational models to optimize architectural geometry based on 30 years of historical EPW data using Ladybug & Honeybee.
 - Projects across Yucatán (Kanasín, Puerto Laguna) and Chile (Pitama — Africam Safari).
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RESEARCH & PUBLICATIONS

La Geométrica Encyclopedia

Founder & Primary Author | 2018 – Present

- The industry's most comprehensive technical reference for 3D geometry and algorithmic logic.
 - Used as the core training system for hundreds of professionals in advanced Rhino/Grasshopper workflows.
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ACADEMIC LEADERSHIP

Dimensión N — Academy of Computational Design

Founder & Director | 2008 – Present

- Established Mexico's premier independent academy for algorithmic design.
- Authorized McNeel (Rhino 3D) Reseller and Trainer.

University Teaching (Select)

- **Tecnológico de Monterrey (ITESM):** Professor of Advanced 3D Modeling for Architecture.
- **Universidad Iberoamericana:** Professor of Rhinoceros & New Technologies. Coordinated the first Digital Fabrication Diploma in Mexico.
- **Universidad Anáhuac:** Professor, Master's in Architecture & Digital Fabrication — Data Visualization.

Advanced Mentorship

- Trained directly by industry pioneers including Raja Issa, Andy Payne, and Chris Mackey.
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TECHNICAL STACK

Software & Logic:

- **Expert:** Rhino 8 (Certified), Grasshopper, Python Scripting, Kangaroo (Physics), Karamba.
- **Proficient:** Maya, Blender, SolidWorks, Adobe Creative Suite.

Reality Capture & AI:

- Drone Photogrammetry, Gaussian Splatting, Point Cloud to Mesh Repair.
- AI Agent Architectures (LLMs, MCP), Three.js, GLSL.

Fabrication Hardware:

- **CNC Milling:** Expert in 2.5, 3, 4 & 5-Axis strategies (Wood, Metal, Foam).
- **Industrial Cutting:** Waterjet & Laser Cutting (Advanced Nesting).
- **Additive Manufacturing:** Management of Large-Format FDM & SLA Resin fleets.
- **Robotics:** KUKA 6-Axis Arm (Workflow Coordination & Production Strategy).

Languages:

- **Spanish:** Native
- **English:** Full Professional Proficiency
- **French:** Conversational

EDUCATION & CREDENTIALS

- **Fab Academy (Candidate)** — MIT Center for Bits and Atoms | Digital Fabrication Focus
 - **Master's in Sustainable Architecture (M.Arch)** — Universidad del Medio Ambiente
 - **Bachelor's in Industrial Design** — Universidad Iberoamericana (2000–2005)
 - **Project Management Diploma (PMI)** — ITAM
 - **Automotive Design Diploma** — Rigoletti Casa de Diseño | Advanced Surface Modeling
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Let's build the future of your project. Reach out for a technical consultation.